Лабораторна робота №13

Семенюка Владислава Григоровича

Група ІПЗ-21

Варіант 7

Код консольної програми

```
using System;
using System.Diagnostics.CodeAnalysis;
using Avalonia.Controls;
using Avalonia.Controls.Templates;
using NBUExchangeApp.ViewModels;

namespace NBUExchangeApp;

/// <summary>
/// Given a view model, returns the corresponding view if
possible.
/// </summary>
[RequiresUnreferencedCode(
    "Default implementation of ViewLocator involves reflection
which may be trimmed away.",
    Url =
"https://docs.avaloniaui.net/docs/concepts/view-locator")]
public class ViewLocator : IDataTemplate
{
    public Control? Build(object? param)
    {
        if (param is null)
            return null;

        var name =
param.GetType().FullName!.Replace("ViewModel", "View",
StringComparison.Ordinal);
        var type = Type.GetType(name);

        if (type != null)
        {
            return (Control)Activator.CreateInstance(type)!;
        }

        return new TextBlock { Text = "Not Found: " + name };
```

```
    }

    public bool Match(object? data)
    {
        return data is ViewModelBase;
    }
}
```

```
macOS, I use Avolania because it's the easiest way to create
UI on macOS
```

```
using Avalonia;
using System;
using NBUExchangeApp.Services;

namespace NBUExchangeApp;

sealed class Program
{
    // Initialization code. Don't use any Avalonia, third-party
APIs or any
    // SynchronizationContext-reliant code before AppMain is
called: things aren't initialized
    // yet and stuff might break.
    [STAThread]
    public static void Main(string[] args) =>
BuildAvaloniaApp()
        .StartWithClassicDesktopLifetime(args);

    // Avalonia configuration, don't remove; also used by
visual designer.
    public static AppBuilder BuildAvaloniaApp()
        => AppBuilder.Configure<App>()
            .UsePlatformDetect()
#if DEBUG
            .WithDeveloperTools()
#endif
            .WithInterFont()
            .LogToTrace();
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<assembly manifestVersion="1.0"
xmlns="urn:schemas-microsoft-com:asm.v1">
```

```
 <!-- This manifest is used on Windows only.
      Don't remove it as it might cause problems with window
transparency and embedded controls.
      For more details visit
https://learn.microsoft.com/en-us/windows/win32/sbscs/applicat
ion-manifests -->
 <assemblyIdentity version="1.0.0.0"
name="NBUExchangeApp.Desktop"/>

 <compatibility
xmlns="urn:schemas-microsoft-com:compatibility.v1">
   <application>
     <!-- A list of the Windows versions that this application
has been tested on
          and is designed to work with. Uncomment the
appropriate elements
          and Windows will automatically select the most
compatible environment. -->

     <!-- Windows 10 -->
     <supportedOS Id="{8e0f7a12-bfb3-4fe8-b9a5-48fd50a15a9a}"
/>
   </application>
 </compatibility>
</assembly>
```

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```

```
using Avalonia;
using Avalonia.Controls.ApplicationLifetimes;
using Avalonia.Markup.Xaml;
using NBUExchangeApp.ViewModels;
using NBUExchangeApp.Views;

namespace NBUExchangeApp;

public partial class App : Application
{
   public override void Initialize()
   {
       AvaloniaXamlLoader.Load(this);
   }

   public override void OnFrameworkInitializationCompleted()
   {
       if (ApplicationLifetime is
IClassicDesktopStyleApplicationLifetime desktop)
       {
           desktop.MainWindow = new MainWindow
           {
               DataContext = new MainViewModel(),
           };
       }

       base.OnFrameworkInitializationCompleted();
   }
}
```

```
<Window xmlns="https://github.com/avaloniaui"
       xmlns:x="http://schemas.microsoft.com/winfx/2006/xaml"
```

```
        xmlns:vm="using:NBUExchangeApp.ViewModels"

xmlns:d="http://schemas.microsoft.com/expression/blend/2008"

xmlns:mc="http://schemas.openxmlformats.org/markup-compatibili
ty/2006"
        xmlns:models="using:NBUExchangeApp.Models"
        mc:Ignorable="d" d:DesignWidth="800"
d:DesignHeight="450"
        x:Class="NBUExchangeApp.Views.MainWindow"
        x:DataType="vm:MainViewModel"
        Icon="/Assets/avalonia-logo.ico"
        Title="NBUExchangeApp">

    <Design.DataContext>
        <vm:MainViewModel />
    </Design.DataContext>


    <Grid Margin="20" RowDefinitions="Auto,Auto,Auto,*,Auto"
RowSpacing="12">

        <TextBlock Grid.Row="0" Text="Курс валют НБУ"
FontSize="24" FontWeight="Bold" />

        <TextBlock Grid.Row="1" Text="Джерело: bank.gov.ua (NBU
Stat Service)" Opacity="0.7" />

        <Button Grid.Row="2" Content="Завантажити з інтернету"
                Command="{Binding LoadCommand}"
Margin="0,16,0,0" />

        <ListBox Grid.Row="3"
                 CornerRadius="12"
                 BorderThickness="0.5"
                 BorderBrush="Gray"
                 Padding="6"
                 ItemsSource="{Binding Readings}"
                 IsVisible="{Binding Readings,
Converter={x:Static ObjectConverters.IsNotNull}}">
            <ListBox.ItemTemplate>
                <DataTemplate
x:DataType="models:ExchangeReading">
```

```
                    <Grid ColumnDefinitions="60,*,180,100"
ColumnSpacing="12">
                        <TextBlock Grid.Column="0"
Text="{Binding Currency}" FontWeight="Bold" />
                        <TextBlock Grid.Column="1"
Text="{Binding Text}" />
                        <TextBlock Grid.Column="2"
                                   Text="{Binding Rate,
StringFormat={}{0:F4} UAH за 1 од.}" />
                        <TextBlock Grid.Column="3"
Text="{Binding ExchangeDate}" Opacity="0.7" />
                    </Grid>
                </DataTemplate>
            </ListBox.ItemTemplate>
        </ListBox>

        <TextBlock Grid.Row="4" Text="{Binding Status}"
FontStyle="Italic" />
    </Grid>


</Window>
```

```
using Avalonia.Controls;

namespace NBUExchangeApp.Views;

public partial class MainWindow : Window
{
    public MainWindow()
    {
        InitializeComponent();
    }
}
```

```
using System;
using System.Collections.Generic;
using System.Threading.Tasks;
using CommunityToolkit.Mvvm.ComponentModel;
using CommunityToolkit.Mvvm.Input;
using NBUExchangeApp.Models;
using NBUExchangeApp.Services;

namespace NBUExchangeApp.ViewModels;
```

```
public partial class MainViewModel : ViewModelBase
{
   [ObservableProperty]
   private string _status = "Натисніть «Завантажити з
інтернету»";

   [ObservableProperty]
   private List<ExchangeReading>? _readings;

   [RelayCommand]
   private async Task LoadAsync()
   {
       Status = "Завантаження…";
       Readings = null;

       try
       {
           List<ExchangeReading> result = await
ExchangeApiService.LoadExchangeDataAsync();
           Readings = result;
           Status = result.Count == 0 ? "Нічого не знайдено" :
$"Дані отримано: {result.Count}";
       }
       catch (Exception)
       {
           Status = "Не вдалося отримати дані";
       }
   }
}
```

```
using CommunityToolkit.Mvvm.ComponentModel;

namespace NBUExchangeApp.ViewModels;

public abstract class ViewModelBase : ObservableObject
{
}
```

```
using System.Collections.Generic;
using System.Linq;
using System.Net.Http;
using System.Text.Json;
using System.Threading.Tasks;
```

```
using  NBUExchangeApp.Models;


namespace NBUExchangeApp.Services;

public class ExchangeApiService
{
   private static readonly HttpClient Client = new
HttpClient();

   const string ACHCodeKey = "r030";
   const string TextKey = "txt";
   const string RateKey = "rate";
   const string CurrencyKey = "cc";
   const string ExchangedateKey = "exchangedate";
   const string SpecialKey = "special";



   public static async Task<List<ExchangeReading>>
LoadExchangeDataAsync()
   {
       const string url =
"https://bank.gov.ua/NBUStatService/v1/statdirectory/exchangen
ew?json";


       // Надсилаємо HTTP GET і отримуємо JSON як текст.
       string json = await Client.GetStringAsync(url);


       // Перетворюємо текст на дерево JSON-елементів.
       using JsonDocument document = JsonDocument.Parse(json);

       var readings = new List<ExchangeReading>();


       foreach (var item in
document.RootElement.EnumerateArray())
       {
           readings.Add(new ExchangeReading
           {
               ACHCode =
item.GetProperty(ACHCodeKey).GetInt32(),
```

```
                Text = item.GetProperty(TextKey).GetString() ??
"Unknown",
                Rate = item.GetProperty(RateKey).GetDouble(),
                Currency =
item.GetProperty(CurrencyKey).GetString() ?? "Unknown",
                ExchangeDate =
item.GetProperty(ExchangedateKey).GetString() ?? "Unknown",
                Special =
item.GetProperty(SpecialKey).GetString() ?? "Unknown"
            } );
        }


        return readings;
    }


}
```

```
namespace NBUExchangeApp.Models;

public class ExchangeReading
{
    public int? ACHCode { get; set; }
    public string? Text { get; set; }
    public double? Rate { get; set; }
    public string? Currency { get; set; }
    public string? ExchangeDate { get; set; }
    public string? Special { get; set; }


}
```

ШІ використовувався тільки для ресьорчу.